Supplementary Information

Pilot-Scale CO_2 Electrolysis Enables a Semi-Empirical Electrolyzer Model

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TABLE OF CONTENTS

Experimental Methods **S3**

MEA Fabrication S3

Lab-scale CO_2 Electrolysis S3

Pilot-scale CO_2 Electrolysis S4

Anodic CO_2 Crossover Measurements S6

Resistance Measurements S7

Anode Kinetic Experiments S7

Electrolyzer Custom Model **S9**

Model Overview S9

CO_2 Mass Transport S10

Anodic CO_2 Crossover Modelling S12

Voltage Distribution S13

Electrochemical Kinetics S13

Pressure and Temperature S17

Water Management S17

Liquid Product Crossover S18

Model Inputs S18

Results and Discussion **S21**

Table S5. Modelling studies of CO_2 electrolysis with gas diffusion electrodes S21

Figure S3. Performance of the lab-scale CO_2 electrolyzer at different cathodic CO_2 supply rates (150 mA cm^{-2}) S22

Figure S4. Performance of the lab-scale CO_2 electrolyzer at different current densities (12 sccm cm^{-2} of cathodic CO_2 supply) S23

Figure S5. Simulated pH profile within the MEA electrolyzer S25

Figure S6. Area-specific ohmic resistance at different temperatures S26

Kinetic Parameter Determination S27

Figure S9. Model validation against 800 cm^2 pilot-scale single cell S30

CO/CO_2 Concentration Discrepancy S31

Figure S11. Model validation against 10 x 800 cm^2 pilot-scale stack S34

References **S35**

EXPERIMENTAL METHODS

MEA Fabrication. The membrane electrode assembly (MEA) in this work consisted of four components: an anode, a spacer, an anion-exchange membrane, and a cathode. Components were layered individually during electrolyzer assembly (e.g. no hot-pressing was involved). A commercial supplier (Magneto Special Anodes BV) provided the catalyst material for the anodic oxygen evolution reaction which was deposited on a porous titanium felt. A hydrophilic spacer material was inserted between the membrane and the anode to reduce the possibility of short-circuits. The anion-exchange membrane (Sustainion X37-50 Grade RT, Dioxide Materials), activated in 1 M potassium hydroxide for at least 24 hours, was washed with deionized water prior to use. For the pilot-scale stack experiment, a different anion-exchange membrane (Ionomr Innovations AF2-CLE8-15-X) was used which was activated in 1 M potassium bicarbonate for at least 24 hours prior to use.

A commercial carbon paper (AvCarb MGL 190) served as the cathode substrate. A microporous layer was added to the carbon paper by blade-coating a slurry across the face of the electrode and treating it in the oven. A catalyst ink comprised of copper nanoparticles and Nafion solution was spray-deposited onto the electrode. A support layer comprised of carbon black and Nafion solution was also spray-deposited onto the electrode.

Lab-scale CO_2 Electrolysis. To train the electrolyzer model, experiments were conducted in a 5 cm^2 CO_2 electrolyzer with a single serpentine channel design for the cathode and anode flow fields (Dioxide Materials). Heating pads on the cathode and anode and a temperature controller were installed to maintain a specified cell temperature of 40 °C.

A thermal mass flow controller (Sierra SmartTrak 100 or Brooks Instruments SLA5850) was used to supply CO_2 at a rate of 60 sccm, unless otherwise specified. The CO_2 was humidified at room temperature (25 °C) before entering the reactor. The cathode outlet was directed into a cold trap (Supelco, Sigma Aldrich) immersed in an ice bath (~ 0 °C) to condense any vapour phase products. The anolyte, 0.1 M potassium bicarbonate (Sigma Aldrich), was recirculated from a 40 mL reservoir into the electrolyzer at a rate of ~10 mL min^{-1} using a peristaltic metering pump (Fisher Scientific).

An electrochemical workstation (Autolab PGSTAT204 with 10 A booster and FRA32 module or Landt Battery Test Station) was used to perform galvanostatic measurements. A single MEA was used to screen multiple currents sequentially, starting at the lowest current. Each current was run for at least 45 minutes to enable the buildup of liquid products. Between each current the cell was turned off, the cathode was washed with ~2 mL of deionized water to remove any salt deposits,[1] and the electrolyte and cold trap liquids were replaced.

Gas samples were taken from the cold trap in 0.5 mL volumes using a gas-tight syringe (Hamilton, Fisher Scientific) after 30 minutes of operation at a specified current. The gas flow rate was measured using a bubble column each time a gas sample was taken. Gas products were quantified using a gas chromatograph (PerkinElmer Clarus 590) equipped with a thermal conductivity detector (TCD) and flame ionization detector (FID). Liquid products from the cold trap and the anolyte were quantified using ^1H nuclear magnetic resonance spectroscopy (NMR) on an Agilent DD2 500 spectrometer with a dimethyl sulfoxide (DMSO) internal standard.

Pilot-scale CO_2 Electrolysis. To validate the electrolyzer model, CO_2 electrolysis was also performed with a custom-made 800 cm^2 electrolyzer and testing system (**Figure S1**). Titanium

machined with a multiple serpentine channel design was used for the cathode and anode flow fields. Experiments were performed with a single cell (800 cm^2) and with a ten-cell stack (8000 cm^2). The electrolyzer temperature of 40 °C was determined by the electrolyte supply temperature, which in turn was controlled by an external heating and cooling loop.

A thermal mass flow controller (Brooks Instruments SLA5861) was used to supply humidified CO_2 at a rate of 0.33 $kgCO_2$ hr^{-1} (single cell) or 3.5 kg CO_2 hr^{-1} (cell stack). CO_2 was humidified before the thermal mass controller in a column of packed beads at elevated temperature and pressure (approximately 51 °C and 53 kPag, respectively). The electrolyte flow rate through the electrolyzer, ~2 L min^{-1} (single cell) or ~20 L min^{-1} (cell stack), was controlled by an electromagnetic flow meter (Endress & Hauser Promag P300, FP3B15).

An industrial power supply (FlexKraft) was used to perform galvanostatic measurements. A single MEA was used to screen multiple currents sequentially, starting at the lowest current. Gas samples were taken after the cell in 1 mL volumes using an autosampler attached to the gas chromatograph (PerkinElmer Clarus 590). The gas chromatograph was equipped with a TCD and FID. The pilot-scale electrolysis system was designed for the operation of multiple electrolyzer stacks at the same time. When operating a single cell or a single ten-cell stack, the gas flow rates could not be accurately quantified within the range of the installed instrumentation. Therefore, it was only possible to measure the stream compositions and not the selectivity. No liquid products were analyzed.

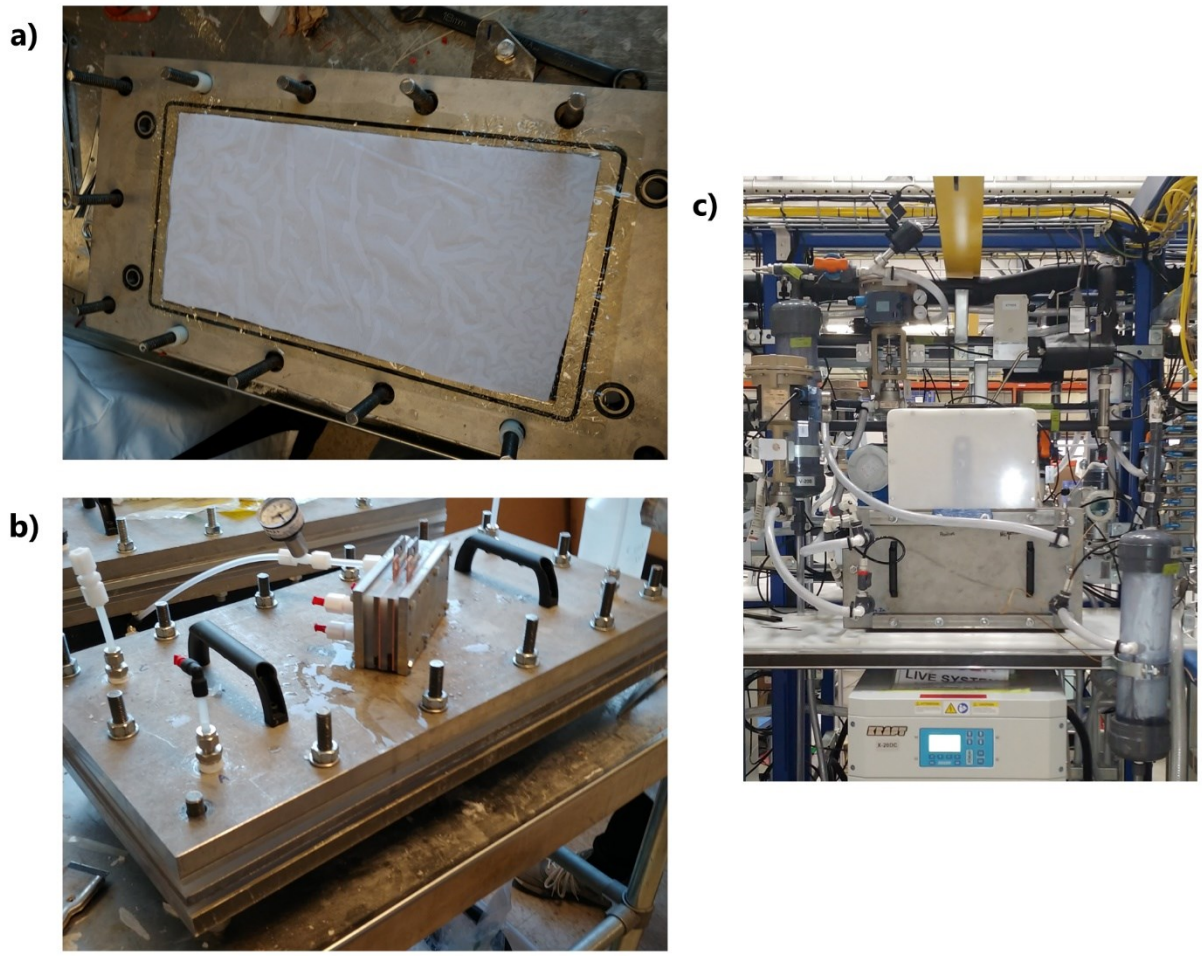

Figure S1. Pilot-scale CO_2 electrolysis hardware. (a) 800 cm^2 pilot-scale cell being assembled for experiment. (b) Benchtop-scale CO_2 electrolyzer (cell active area of 25 cm^2) placed on top of pilot-scale CO_2 electrolyzer (cell active area of 800 cm^2). (c) Pilot-scale CO_2 electrolysis stack in operation.

Anodic CO_2 Crossover Measurements. The anolyte of the MEA-based CO_2 electrolyzer was stored in a sealed reservoir with an exhaust port for the anode gas to escape. The anode gas flow rate was measured in an identical manner to that of the cathode gas described in the main text (e.g., using bubble column measurements at ambient temperature and pressure). The anode gas was

assumed to be comprised of carbon dioxide (CO_2) and oxygen (O_2), exclusively. It was assumed that there was negligible humidity in the stream and that no gaseous products of CO_2 electrolysis could travel through the anion exchange membrane. The molar flow rate of O_2 was calculated based on the electrolyzer current and the O_2 Faradaic Efficiency (FE). The FE towards O_2 was assumed to be equivalent to the complement of the total cathodic FE (e.g., the sum of hydrogen (H_2) and CO_2 electrolysis products). For example, if the CO_2 electrolysis products combined with H_2 embodied 95% of the reaction current, it was assumed that the remaining 5% of the current went into liquid products which migrated through the membrane and were oxidized back to CO_2 on the anode. In this case, the anodic FE towards O_2 evolution was assumed to be 95% with 5% attributed to liquid product oxidation. If the total cathodic FE exceeded 100%, the anodic FE for O_2 evolution was assumed to be 100%.

Resistance Measurements. The ohmic resistance was determined using Electrochemical Impedance Spectroscopy (EIS). EIS was performed between 2 kHz and 100 kHz with 10 points per decade and an amplitude of 0.0001 A. The EIS was performed after operation galvanostatic operation at a specific current density (25, 50, or 75 mA cm^{-2}). A handheld multi-meter was used to determine the ohmic resistance of the interconnects by measuring the applied voltage at the cell at a series of current densities and comparing it to the galvanostat output. The interconnect resistance was deducted from the EIS-measured resistance (high frequency intercept of the Nyquist plot) and then multiplied by the cell area (5 cm^2) to obtain the area specific cell resistance.

Anode Kinetic Experiments. The anode kinetic constants were derived based on experiments in a three-electrode setup. The working electrode (10 cm^2 of titanium felt), a carbon paper counter electrode, and an Ag/AgCl reference electrode (1 M potassium chloride) were immersed in 0.5 M potassium bicarbonate ($KHCO_3$). A higher concentration of electrolyte was used for this

experiment to increase the electrolyte conductivity and lower the ohmic losses to an acceptable level for measurement. CO_2 was bubbled through the electrolyte to achieve saturation prior to starting the experiment. The beaker with the three electrodes was placed on a hot plate to achieve a temperature of 40 °C in the stirred electrolyte (1200 rpm). Galvanostatic electrolysis was performed at 12 different current densities with each step lasting for 5 minutes. The measured potentials were iR-compensated and converted to the Reversible Hydrogen Electrode (RHE) using the bulk electrolyte pH (measured at 6.90 prior to the experiment). The reference electrode potential and thermodynamic voltage were calculated at 40 °C using literature values.[2,3]

ELECTROLYZER CUSTOM MODEL

Model Overview. In the model, humidified CO_2 (stream 1, **Figure S2**) and anolyte (stream 4, **Figure S2**) are supplied to the cathode and anode, respectively. Although it is rare for MEA-based CO_2 electrolyzers to operate on pure water anolyte, the model assumes the anolyte to be pure water and any chemical reactions with the electrolyte (e.g., salt precipitation) are ignored. The CO_2 reduction reaction (CO_2RR) and hydrogen (H_2) evolution reaction take place on the cathode while the O_2 evolution reaction (OER) takes place on the anode. Unreacted CO_2, water vapor, H_2, and gas products of CO_2RR – carbon monoxide (CO) and C_2H_4 – leave the cathode (stream 2, **Figure S2**) mixed with aqueous solutions of CO_2RR liquid products – formate ($HCOO^-$), acetate (CH_3COO^-), ethanol (C_2H_5OH), and 1-propanol (C_3H_7OH) (stream 3, **Figure S2**). A humid gas stream of CO_2 and O_2 leave the anode (stream 5, **Figure S2**) as does a liquid stream of unused water from the anolyte and liquid products of CO_2RR liquid products which cross through the membrane (stream 6, **Figure S2**).

There are four primary modules of the electrolyzer model each with numerous interdependencies. The first module, CO_2 mass transport, is used to predict the CO_2 concentration at the catalyst surface as a function of mass transport effectiveness within the electrolyzer and bulk CO_2 concentration. The second module, CO_2 crossover, estimates the amount of CO_2 that will be generated on the anode via carbonate/bicarbonate transport through the membrane and liquid product oxidation. The third module, voltage distribution, divides the applied cell voltage between ohmic losses, Nernstian losses, and electrode potentials. The fourth module, electrochemical kinetics, predicts the reaction rate for cathodic and anodic reactions based on the CO_2 availability and the electrode potentials. There are three secondary modules which deal with temperature and pressure, water management, and liquid product crossover.

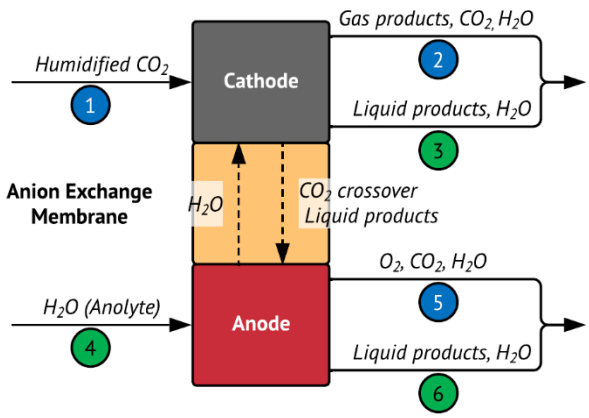

Figure S2. Depiction of fluid flows within the MEA electrolyzer. Gas products include H_2, CO, and C_2H_4 while liquid products include $HCOO^-$, CH_3COO^-, C_2H_5OH, and C_3H_7OH. The blue and green circles depict gas and liquid phase streams, respectively.

CO_2 Mass Transport. CO_2 is supplied to the electrolyzer is assumed to be equally divided amongst all cells in the stack. CO_2 in the channel must migrate to the catalyst surface to be electrochemically converted to products or chemically converted to (bi)carbonates. The molar flux of CO_2 towards the catalyst surface, $\dot{N}_{CO_2}$, is then:

$$\dot{N}_{CO_2} = \sum \frac{a_i j_i}{b_i F} + \tau j_T \tag{S1}$$

where a_i denotes the number of carbon molecules in one molecule of product i, j_i denotes the partial current towards product i, b_i denotes the number of electrons required for each product molecule i, F denotes Faraday's constant, τ denotes the anodic CO_2 coefficient, and j_T denotes the total current density. The molar flux to the catalyst surface results in a concentration gradient:

$$\dot{N}_{CO_2} = k_{Net}\left(C_{CO_2\,Bulk} - C_{CO_2\,Surf}\right) \tag{S2}$$

where k_{Net} denotes the net mass transfer coefficient, $C_{CO_2\,Bulk}$ denotes the CO_2 concentration in the flow channel, $C_{CO_2\,Surf}$ denotes the CO_2 concentration at the catalyst surface. To account for CO_2 depletion in the electrolyzer, $C_{CO_2\,Bulk}$ is taken to be the average of the inlet and outlet CO_2 concentrations, $C_{CO_2}^{In}$ and $C_{CO_2}^{Out}$, respectively:

$$C_{CO_2\,Bulk} = \frac{1}{2}\left(C_{CO_2}^{In} + C_{CO_2}^{Out}\right) \tag{S3}$$

The mass transfer coefficient is a measure of how effectively CO_2 can be replaced at the catalyst surface. In this model there are two sources of mass transfer resistance considered in series: the serpentine channels and the cathode electrode structure. The net mass transfer coefficient is calculated as follows:

$$\frac{1}{k_{Net}} = \frac{1}{k_S} + \frac{1}{k_E} \tag{S4}$$

where k_S denotes the mass transfer coefficient from the bulk gaseous flow to the surface of the electrode and k_E denotes the mass transfer coefficient through the cathode electrode structure. The Sherwood number (*Sh*) is used to calculate the mass transfer coefficient within the serpentine channels:

$$\theta Sh = \frac{k_S L_c}{D_{CO_2}} \tag{S5}$$

where Θ denotes an empirical mass transfer coefficient modifier, L_c denotes the critical length or hydraulic diameter of the channel, and D_{CO_2} denotes the diffusivity of CO_2 in the gaseous cathode stream. Treating the humidified CO_2 stream as pure CO_2 at 40 °C, the Reynolds numbers

inside the cathode flow fields of the lab-scale cell, pilot-scale cell, and pilot-scale stack were calculated as 140, 630, and 680, respectively. Flow through the cathode serpentine channels is thus assumed to be laminar in this model, as in most fuel cell literature.[4] Exact values of the Nusselt number, the heat-transfer analogue of the Sherwood number, are readily available for laminar flow through rectangular ducts.[5] Since the serpentine channels do not reach the entire active area, e.g., there is some land area, then an empirical mass transfer coefficient (assumed to be between 0 and 100 %) was added to the model. Nominally, if half of the geometric area was land area then the empirical mass transfer coefficient would be 50% if there was no mixing of the cathode streams under the land area but in reality there may be some if the channel geometry is well-designed.[6]

Applying a Bruggeman model effective diffusivity to the cathode electrode structure,[7,8] the mass transfer coefficient is then:

$$k_E = \frac{\varepsilon_E^{1.5}}{t_E} D_{CO_2} \tag{S6}$$

where ε_E denotes the porosity of and t_E denotes the thickness of the electrode.

Anodic CO_2 Crossover Modelling. In aqueous environments, dissolved CO_2 can react in a series of chemical equilibria:

$$CO_2 + OH^- \rightleftharpoons HCO_3{}^- \tag{S7}$$

$$HCO_3{}^- + OH^- \rightleftharpoons CO_3{}^{2-} + H_2O \tag{S8}$$

Depending on the charge carrier through the membrane, there will be a different amount of CO_2 content in the anode gas stream, denoted by the variable τ. To simplify the complex set of equilibria

shown in (S7) and (S8), the anodic CO_2 coefficient is assumed to be a separable function dependent on two variables, $C_{CO_2\ Bulk}$ and j_T:

$$\tau = f\left(C_{CO_2\ Bulk}\right) \times g(j_T) \tag{S9}$$

Voltage Distribution. The voltage applied to the cell, V_{Cell}, is divided amongst several sources, as shown in the following equation:

$$V_{Cell} = E_C - E_A - \Delta V_{Ner} - j_T R_{Ohm} \tag{S10}$$

where E_C denotes the cathode half-cell potential expressed *vs.* the reversible hydrogen electrode (RHE), E_A denotes the anode half-cell potential expressed *vs.* RHE, ΔV_{Ner} denotes the thermodynamic voltage drop due to differences in cathode and anode pH, and R_{Ohm} denotes the area-specific ohmic resistance in the cell.

The Nernstian losses associated with the non-standard pH at the cathode and anode is calculated as follows:

$$\Delta V_{Ner} = \frac{RT}{F log_{10} e}\left(pH_{Cathode} - pH_{Anode}\right) \tag{S11}$$

where R denotes the universal gas constant, T denotes the absolute cell temperature, F denotes Faraday's constant, $pH_{Cathode}$ denotes the local cathode pH, and pH_{Anode} denotes the local anode pH. For this work the cathode and anode pH were derived using COMSOL simulations.

Electrochemical Kinetics. The general form of the CO_2RR can be written as follows for basic and acidic media, respectively:

$$aCO_2 + be^- + cH_2O \rightarrow C_aH_{2c-b}O_{2a+c-b} + bOH^- \tag{S12}$$

$$aCO_2 + be^- + bH^+ \rightarrow C_aH_{2c-b}O_{2a+c-b} + (b-c)H_2O \quad \text{(S13)}$$

where c_i denotes the number of water molecules consumed (at the cathode) to form one molecule of product i under alkaline conditions. The model considers the generation of H_2, two gas-phase CO_2RR products, and four liquid-phase CO_2RR products (**Table S1**).

Table S1. Stoichiometric coefficients for the cathodic reactions in the electrolyzer model. *Based on the cathode pH it is most likely that formic acid and acetic acid are in the form of their conjugate base – $HCOO^-$ and CH_3COO^-, respectively.

Product	**Chemical Formula**	**State at SATP**	**a**	**b**	**c**
Hydrogen	H_2	Gas	0	2	2
Carbon monoxide	CO	Gas	1	2	1
Formic acid*	HCOOH*	Liquid	1	2	2
Acetic acid*	CH_3COOH*	Liquid	2	8	6
Ethylene	C_2H_4	Gas	2	12	8
Ethanol	C_2H_5OH	Liquid	2	12	9
1-Propanol	C_3H_7OH	Liquid	3	18	13

The partial current density towards product i (j_i) is calculated using Tafel kinetics with added CO_2 concentration considerations and pH effects neglected:[7–10]

$$j_i = j_{o,i}\left(\frac{C_{CO_2\,Surf}}{C_{CO_2\,Ref}}\right)^{a_i} exp\left(\eta_i \frac{\alpha_i F}{RT}\right) \quad \text{(S14)}$$

where $j_{o,i}$ denotes the exchange current density for product i, $C_{CO_2\,Ref}$ denotes a reference CO_2 concentration, η_i denotes the cathode overpotential for product i, and α_i denotes the dimensionless transfer coefficient for product i. The exchange current density and the transfer coefficient are

catalyst-specific kinetic parameters. The cathode overpotential for product i, denoted η_i, is calculated as follows:

$$\eta_i = E_C - E_{td,i} \tag{S15}$$

where $E_{td,i}$ is the thermodynamic reduction potential for product i. The thermodynamic reduction potential is calculated using the Gibb's free energy change of the acidic CO_2RR reaction (S13):

$$-E_{td,i} b_i F = (b_i - c_i) G_{H_2O} + G_i - a_i G_{CO_2} \tag{S16}$$

where G_{H_2O} denotes the Gibb's free energy of formation for liquid H_2O, G_i denotes the Gibb's free energy of formation for product i (in whatever state the product is found at SATP), and G_{CO_2}denotes the Gibb's free energy of formation for gaseous CO_2. The Gibb's free energy of formation is calculated using the Aspen properties of the fluid at the specified reactor temperature. The electrolyzer model uses the thermodynamic properties for formic acid and acetic acid rather than their conjugate bases ($HCOO^-$ and CH_3COO^-, respectively). This does introduce some slight error in the model as the dissociation of these acids will result in a different pH-dependent thermodynamic potential. However, the generation rate of these two products is relatively minor, so the impacts of this approximation are likely to be small.

The total current density of the reactor, denoted j_T, is the sum of the partial current densities of each of the cathodic products:

$$j_T = \sum j_i \tag{S17}$$

where j_i denotes the partial current density towards product i. The Faradaic efficiency towards a specific cathodic product i, denoted FE_i, is calculated in the following manner:

$$FE_i = \frac{j_i}{j_T} \tag{S18}$$

The model assumes that OER takes place on the anode with 100% Faradaic efficiency. In otherwords, the partial current of O_2 evolution is equivalent to the total current density of the reactor. O_2 evolution can be written as follows for basic and acidic media, respectively:

$$4OH^- \rightarrow 2H_2O + 4e^- + O_2 \tag{S19}$$

$$2H_2O \rightarrow 4H^+ + 4e^- + O_2 \tag{S20}$$

The OER kinetics take place as per the following reaction:

$$j_T = j_{o,OER} exp\left(\eta_{OER} \frac{\alpha_{OER} F}{RT}\right) \tag{S21}$$

where $j_{o,OER}$ denotes the OER exchange current density, η_{OER} denotes the OER overpotential, and α_{OER} denotes the transfer coefficient for OER. The overpotential for O_2 evolution, denoted η_{OER}, is calculated as follows:

$$\eta_{OER} = E_{td,OER} - E_A \tag{S22}$$

where E_A denotes the anode half-cell potential expressed *vs.* RHE and $E_{td,OER}$ denotes the thermodynamic potential for O_2 evolution. The thermodynamic oxidation potential is calculated using the Gibb's free energy change of the acidic anode reaction (S20):

$$E_{td,OER} 4F = G_{H_2O} - G_{O_2} \tag{S23}$$

where 4 is the number of electrons transferred per molecule of O_2 produced, F denotes Faraday's constant, G_{H_2O} denotes the Gibb's free energy of formation for liquid H_2O, and G_{O_2} denotes the Gibb's free energy of formation for gaseous O_2. The Gibb's free energy of formation is calculated using the Aspen properties of the fluid at the specified reactor temperature.

The heat flux generated by the reactor, denoted $Q_{Reactor}$, is dependent on the thermoneutral voltage. If the cell voltage, denoted V_{Cell}, is above the thermoneutral voltage, then any additional voltage is expelled as heat:

$$Q_{Reactor} = j_T\left(E_{tn,cell} - V_{Cell}\right) \tag{S24}$$

where $E_{tn,cell}$ denotes the thermoneutral voltage of the cell. Here a positive value for heat is expected, indicative that heat is leaving the reactor. The thermoneutral voltage of the cell is calculated based on the Faradaic efficiencies of the different cathodic products:

$$E_{tn,cell} = \sum FE_i\, E_{tn,i} \tag{S25}$$

where $E_{tn,i}$ denotes the thermoneutral voltage of product i. The thermoneutral voltage of a specific cathodic product is based on the enthalpy change of the associated full cell reaction:

$$-E_{tn,i} b_i F = H_i + \frac{1}{4} b_i H_{O_2} - a_i H_{CO_2} - \left(c_i - \frac{1}{2} b_i\right) H_{H_2O} \tag{S26}$$

where H_i denotes the enthalpy of formation of product i (in the appropriate physical state at SATP), H_{O_2} denotes the enthalpy of formation of gaseous O_2, H_{CO_2}denotes the enthalpy of formation of gaseous CO_2, and H_{H_2O} denotes the enthalpy of formation of liquid water. The enthalpies of formation are calculated using Aspen properties at the reactor temperature.

Pressure and Temperature. The electrolyzer is assumed to operate at a constant pressure (e.g., pressure drop through the electrolyzer is neglected). The cathode and anode pressure need not be the same as they are separate variables in the model. Electrolysis is assumed to take place isothermally, with any heat generated from the reaction transferred to the surroundings and not into the exhaust streams. The temperature of the electrolyzer, used for all chemical and electrochemical property calculations, is determined by the temperature of the incoming anolyte, since it likely has a higher thermal inertia than the incoming gas stream.

Water Management. Water plays an important role in CO_2RR: water is supplied to the cathode and anode in vapour and liquid phase, respectively. During the reaction it is consumed by the cathodic reaction and produced by the anodic reaction, combining these reactions into one overall

full cell reaction shows that for all the products (except CO) the reaction will consume water. Water also leaves the reactor as liquid and as vapour in four different streams.

Water leaves the reactor in two liquid streams and two gas streams (four in total). The cathode and anode tail gas streams are assumed to have the same relative humidity. There is water crossover through the anion exchange membrane from the anode to the cathode via diffusion.[11,12] In this model, the molar flux of water crossover through the membrane, denoted $\dot{N}_{H2O\ crossover}$, was assumed to be proportional to the cathodic water consumption rate:

$$\dot{N}_{H2O\ crossover} = \theta_{Water} \sum \frac{c_i j_i}{b_i F} \tag{S27}$$

where θ_{Water} denotes the empirical water crossover coefficient of the membrane.

Liquid Product Crossover. Liquid CO_2RR products produced on the cathode of MEA electrolyzers can migrate through the anion exchange membrane to the anode.[13] Some of the liquid products are negatively charged, facilitating easy migration through anion exchange membranes whereas other liquid products without a charge may be dragged through with other anions.[14,15] A constant anodic crossover fraction, denoted θ_{Liquid}, is assumed by the model for all liquid products.

Model Inputs. It was necessary to supply certain experimental details to the model so that it could understand the situation. Parameters pertaining to the electrolyzer geometry, cathode supply, anode supply, and operating conditions are provided in **Table S2** below. Using lab-scale experiments and simulations in conjunction with the model, certain fitting factors (**Table S3**) and kinetic constants (**Table S4**) were then obtained.

Table S2. Model inputs for lab-scale and pilot-scale CO_2 electrolysis.

Variable Name	Description	Lab-scale	Pilot-scale (Cell)	Pilot-scale (Stack)
wc	Width of flow channel (m)	8.0E-4	1.63E-3	1.63E-3
ec	Depth of flow channel (m)	8.0E-4	5.0E-3	5.0E-3
A_geometric	Active area (m^2)	5.0E-4	8.0E-2	8.0E-1
Sherwood	Sherwood number	2.98	3.96	3.96
theta_channel	Empirical mass transfer coefficient	0.50		
epsilon_e	Cathode GDE porosity	0.78		
thick_e	Cathode GDE thickness (m)	2.0E-4		
P_cathode	Cathode pressure (Pa)	1.01E5		
P_anode	Anode pressure (Pa)	1.01E5		
T	Stack temperature (K)	313		
humidity	Relative humidity of cathode and anode tail gas	01.00		
CCO2_ref	CO_2 reference concentration (kmol m^{-3})	1E-3		
theta_water	Fraction of required cathodic water which crosses over the membrane	1.5		
theta_liquid	Fraction of liquid products which crossover the membrane	0.90		
<F (Cathode)	Molar supply rate to cathode (kmol s^{-1})	4.22E-8	2.26E-6	2.41E-5
<z("CO2")	Molar fraction of incoming CO_2	0.969	0.915	0.918
<z("H2O")	Molar fraction of incoming H_2O	0.0313	0.0854	0.0818
F_anode	Molar supply rate to anode (kmol s^{-1})	9.25E-6	1.85E-3	1.85E-2
<z("H2O")	Molar fraction of incoming H_2O	1.00		

Table S3. Model fitting parameters derived from lab-scale experiments and simulations. For more information on their derivation, please refer to **Figure 2** in the main text.

Variable Name	**Description**	**Value**
carb1	Carbonate-current y-intercept (μmol C^{-1})	7.87
carb2	Carbonate-current slope (μmol cm^2 C^{-1} mA^{-1})	-0.011
V_nernstian	Nernstian voltage loss (V)	-0.51
ohmic1	Ohmic resistance pre-exponential constant (Ω cm^2)	17.5
ohmic2	Ohmic resistance exponential factor (cm^2 mA^{-1})	-0.198

Table S4. Electrochemical kinetic constants derived from lab-scale experiments. For more information on their derivation, see the **Kinetic Parameter Determination** section below. The partial current density towards CH_3COO^- was too low for an accurate kinetic analysis.

Product	**Transfer Coefficient** α_i	**Exchange Current Density** $j_{o,i}$ **(mA** cm^{-2}**)**
O_2	0.222	3.51E-1
H_2	0.256	2.51E-1
CO	0.264	3.57E-2
C_2H_4	0.838	2.72E-10
$HCOO^-$	0.336	8.03E-3
C_2H_5OH	0.476	1.36E-7
C_3H_7OH	0.541	3.86E-10
CH_3COO^-	N/A	N/A

RESULTS AND DISCUSSION

Table S5. Modelling studies of CO_2 electrolysis with gas diffusion electrodes.

Reference	**Products**	**Geometric Area**	**Dimensionality**
Löffelholz et al.[16]	CO, H_2	10 cm^2	Pseudo 2D
Heßelmann et al.[17]	CO, H_2	10 cm^2	1D
Yang et al.[18]	$HCOO^-$, CO, H_2	2 – 40 cm^2	2D
Subramanian et al.[19]	CO, H_2	5 cm^2	3D
Nesbitt and Smith[20]	CO, H_2	1 cm^2	1D
McCallum et al.[8]	CO, C_2H_4, H_2, C_2H_5OH	5 cm^2	1D
Kas et al.[21]	CO, H_2	1 cm^2	2D
Gabardo et al.[22]	CO, C_2H_4, H_2	5 cm^2	1D
Weng et al.[7]	CO, H_2	0.075 cm^2	1D
Wu et al.[23]	CO, H_2	1 cm^2	2D

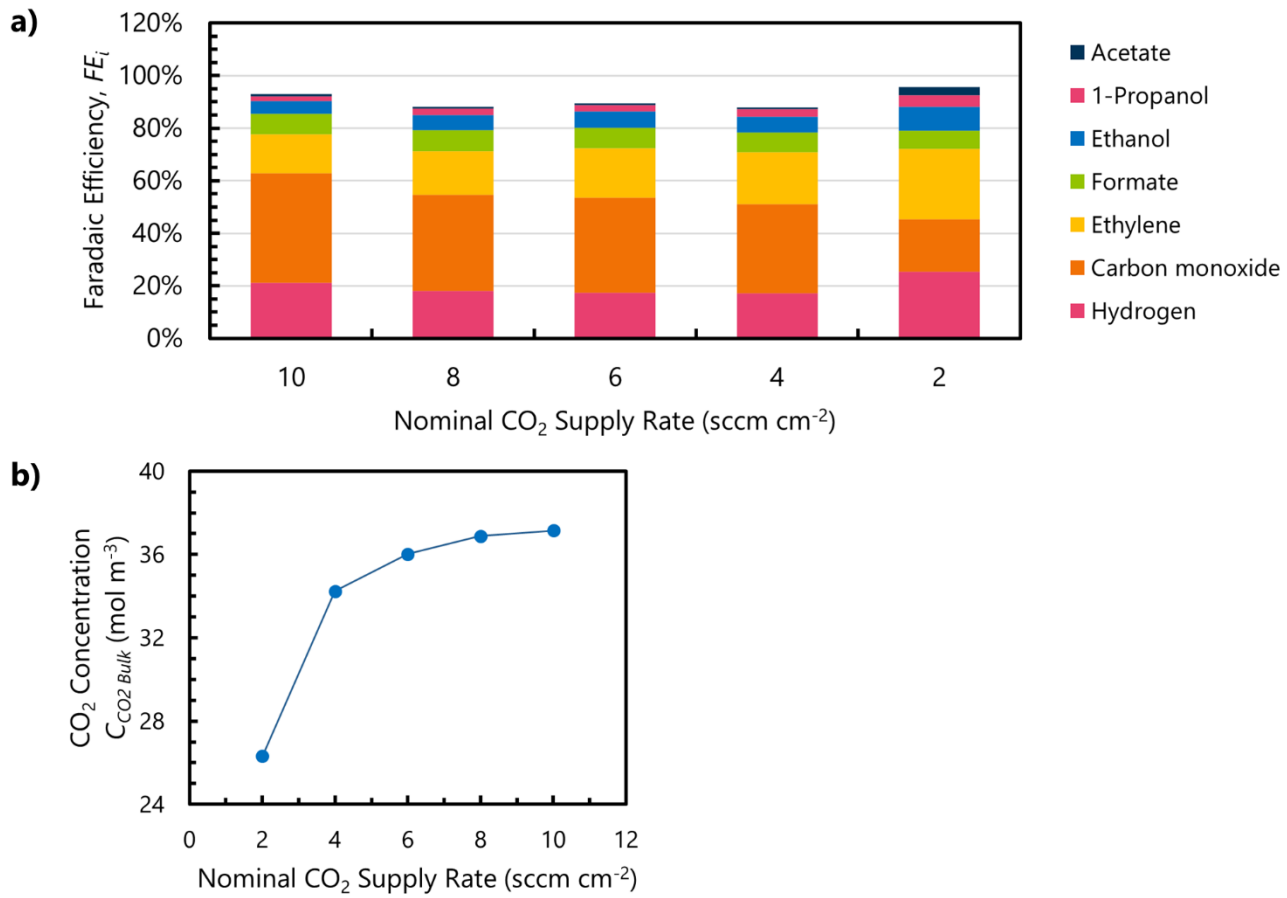

Figure S3. Performance of the lab-scale CO_2 electrolyzer at different cathodic CO_2 supply rates (150 mA cm^{-2}). (a) Faradaic efficiency. Trace amounts (<<1% FE) of additional gas (methane) and liquid (propionaldehyde and acetaldehyde) products were detected but were omitted from the model. (b) Cathodic CO_2 concentration predicted by the electrolyzer model. Error bars correspond to the standard deviation of three independent measurements.

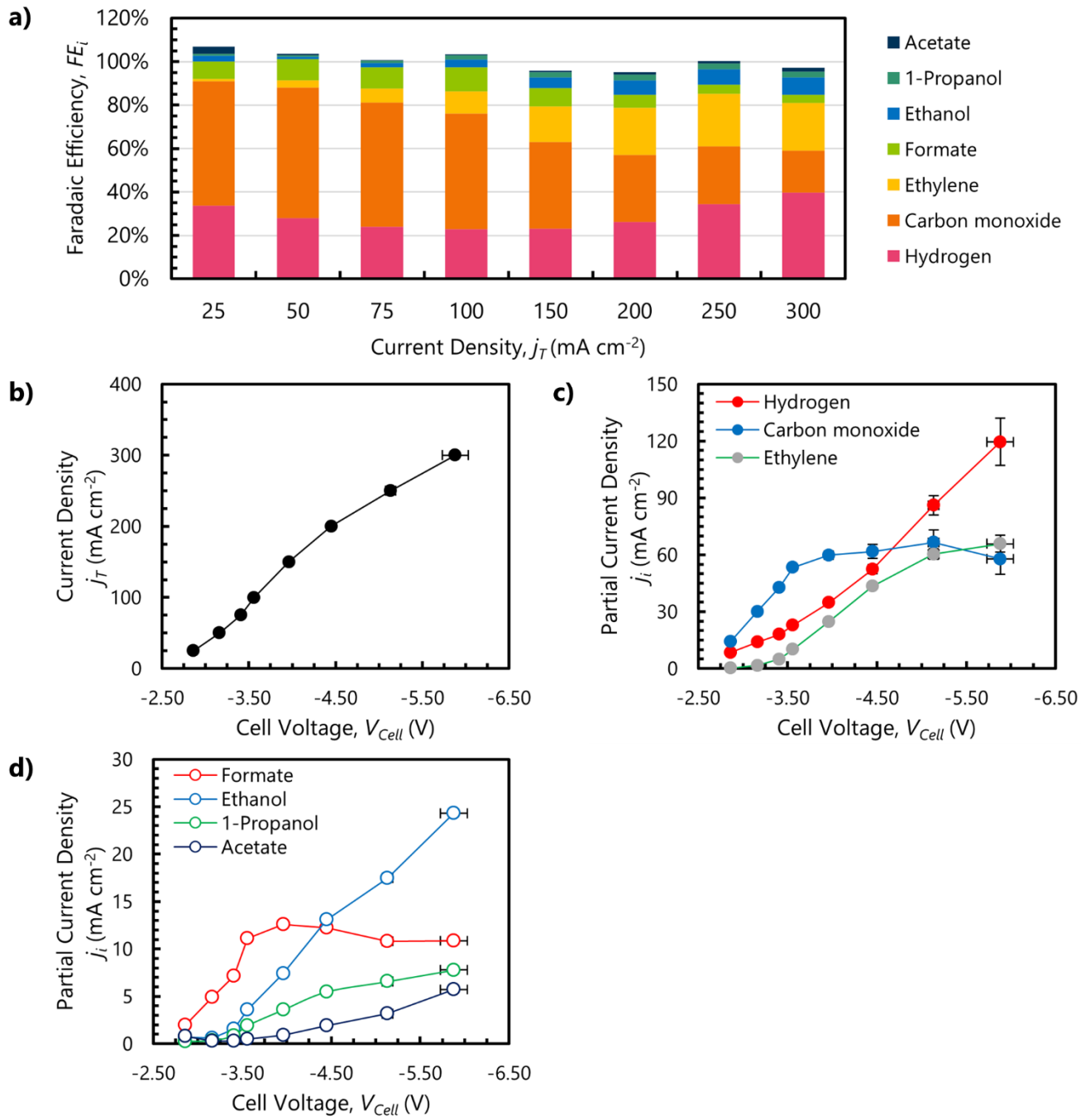

Figure S4. Performance of the lab-scale CO_2 electrolyzer at different current densities (12 sccm cm^{-2} of cathodic CO_2 supply). (a) Faradaic efficiency. Trace amounts (<<1% FE) of additional gas (methane) and liquid (propionaldehyde and acetaldehyde) were detected but were omitted from the model. (b) Cell voltage. (c) Partial current density towards gas products at different cell

voltages. (d) Partial current density towards liquid products at different cell voltages. Error bars on the cell voltage correspond to the standard deviation of the cell voltage over the final 15 minutes of operation. Other error bars correspond to the standard deviation of three independent measurements.

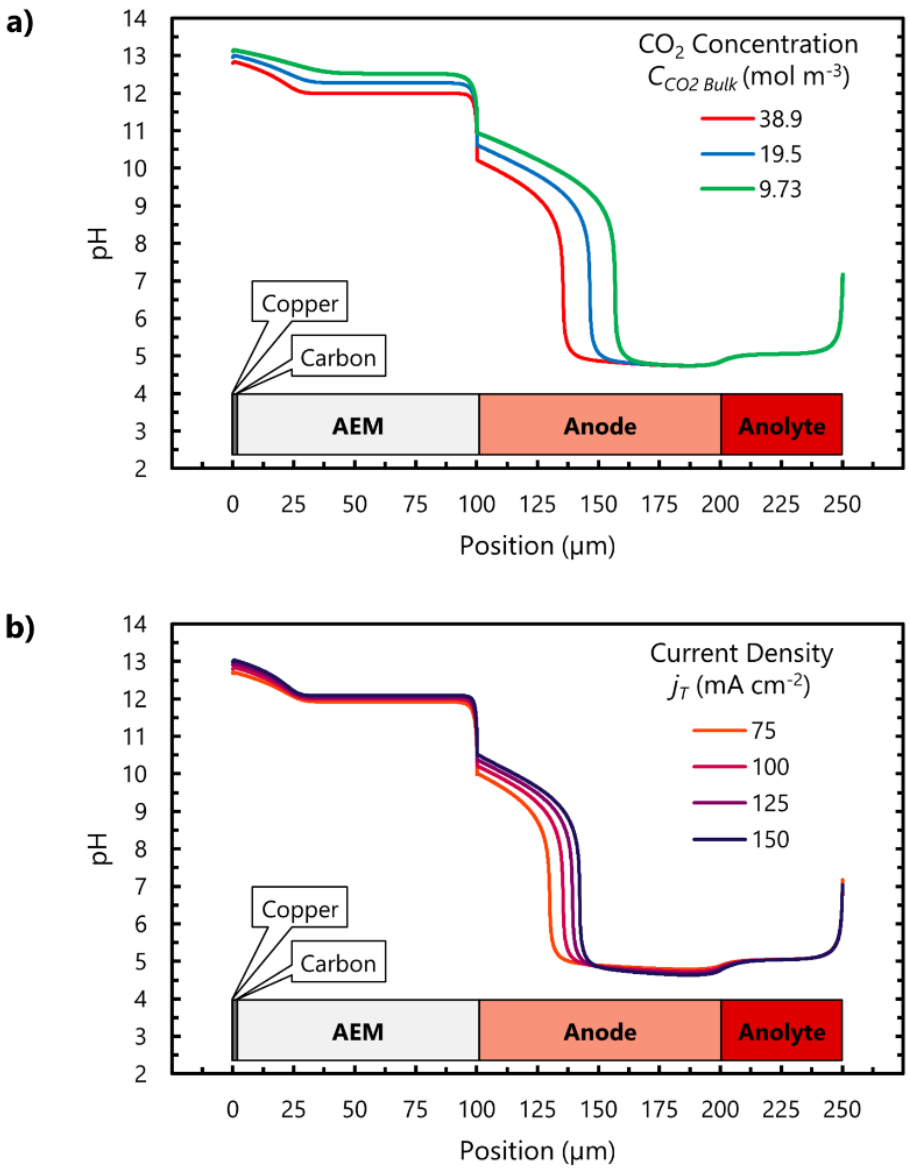

Figure S5. Simulated pH profile within the MEA electrolyzer. (a) The pH profile at different bulk CO_2 concentrations. (b) The pH profile at different current densities. The simulated MEA consisted of five layers: a copper catalyst layer (150 nm thick), a carbon support layer (2 μm thick), an anion exchange membrane (AEM, 100 μm thick), a porous anode layer (100 μm thick), and an anolyte layer (0.1 M $KHCO_3$, 50 μm thick).

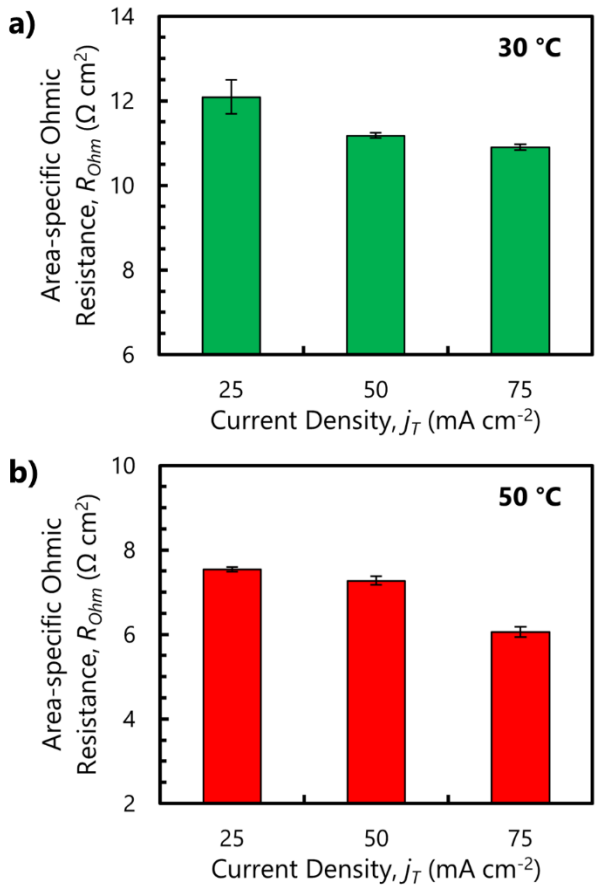

Figure S6. Area-specific ohmic resistance at different temperatures. (a) 30 °C. (b) 50 °C. Error bars correspond to the standard deviation of two independent measurements.

Kinetic Parameter Determination. To quantify the anode kinetic constants, a series of ex-situ experiments were conducted in near-neutral anolyte (see **Anode Kinetic Experiments** section above). Consistent with other reports of iridium-based catalysts, there was a larger transfer coefficient in low potential regions than in high potential regions, suggesting a change in the rate-determining step and slower kinetics (**Figure S7a**).[24,25] The OER kinetic constants (**Table S4**) were derived from the high current density regime (e.g., 25-150 mA cm^{-2}, solid markers in **Figure S7a**). The overpotentials in this region are comparable to other studies using near-neutral electrolytes which have demonstrated worse kinetics than in extreme pH electrolytes (e.g., alkaline or acidic).[22,26,27]

To determine the cathode kinetic constants, (S14) was rearranged as follows:

$$log_{10}\left(j_i\left(\frac{C_{CO_2\,Surf}}{C_{CO_2\,Ref}}\right)^{-a_i}\right) = log_{10}j_{o,i} + \alpha_i\left(\frac{\eta_i F}{RT}log_{10}\,e\right) \tag{S28}$$

where the term on the left side of the equation describes a scaled current density, $\acute{j}$, and the term at the furthest right describes a scaled overpotential, $\acute{\eta}$. Plotting the scaled current density against the scaled overpotential using different experimental data points was expected to yield a linear relationship wherein the intercept and the slope of trendline can be used to derive the exchange current density and activity coefficient, respectively.

To quantify the cathode kinetic constants, results from the previous experiments (**Figure S4**) were integrated with the electrolyzer model. The model was necessary to estimate the surface CO_2 concentration since it was not measured experimentally. The anode kinetic constants, Nernstian voltage losses, and ohmic resistances were also integrated to estimate the cathode potential at each applied cell voltage. Electrolysis at 200, 250, and 300 mA cm^{-2} were excluded from the kinetic analysis due to the uncertainty in extrapolating the ohmic loss function to such high currents (gas

products in **Figure S7**, liquid products in **Figure S8**, and product-specific kinetic constants in **Table S4**).

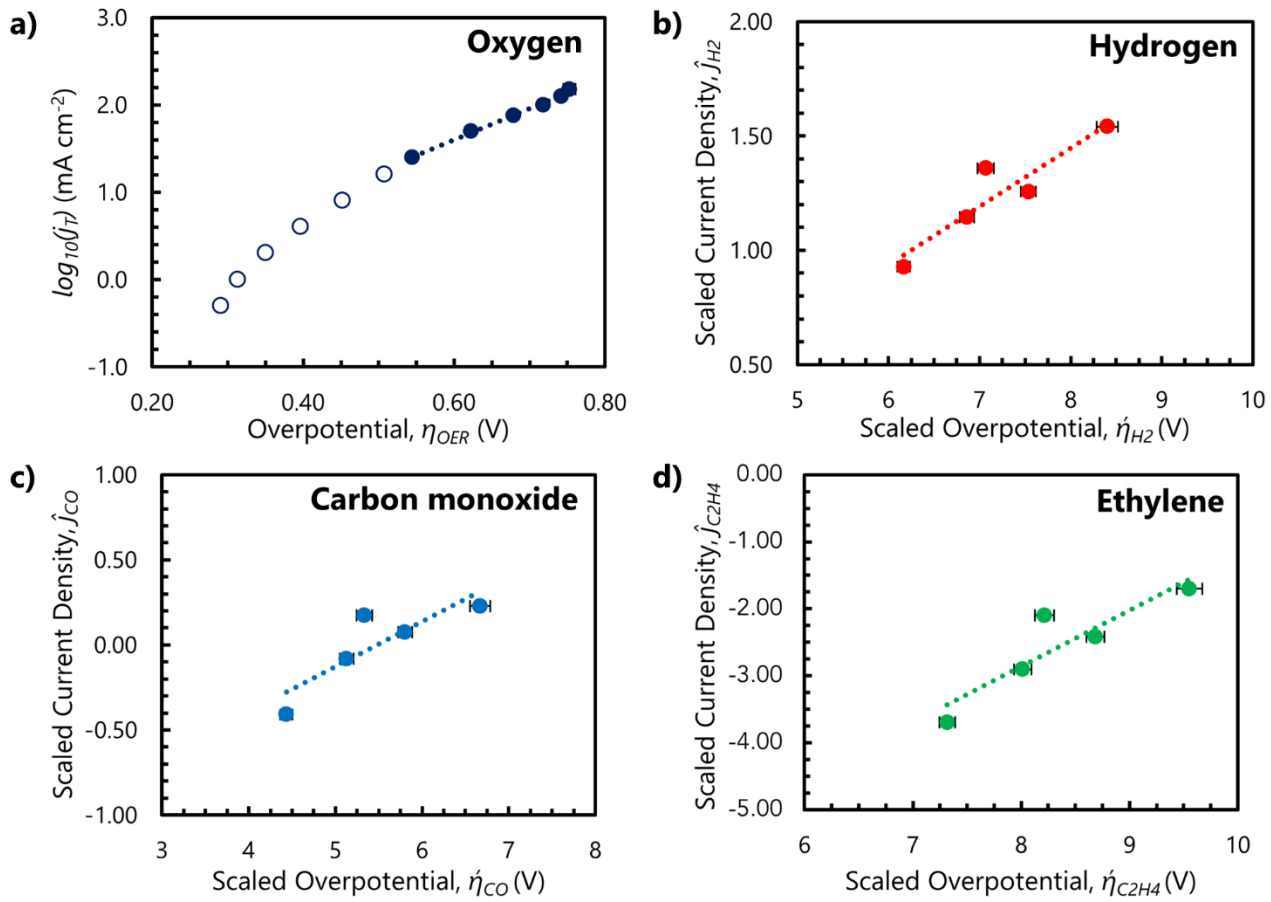

Figure S7. Electrochemical kinetics for gas products in the CO_2 electrolyzer. (a) Oxygen. (b) Hydrogen. (c) Carbon monoxide. (d) Ethylene. The values in (b), (c), and (d) were derived, using the data from CO_2 electrolysis at five different currents: 25, 50, 75, 100, 150 mA cm^{-2}. Horizontal error bars correspond to the standard deviation of the cell voltage over the final 15 minutes of operation. Vertical error bars in (b), (c), and (d) correspond to the standard deviation of three independent measurements.

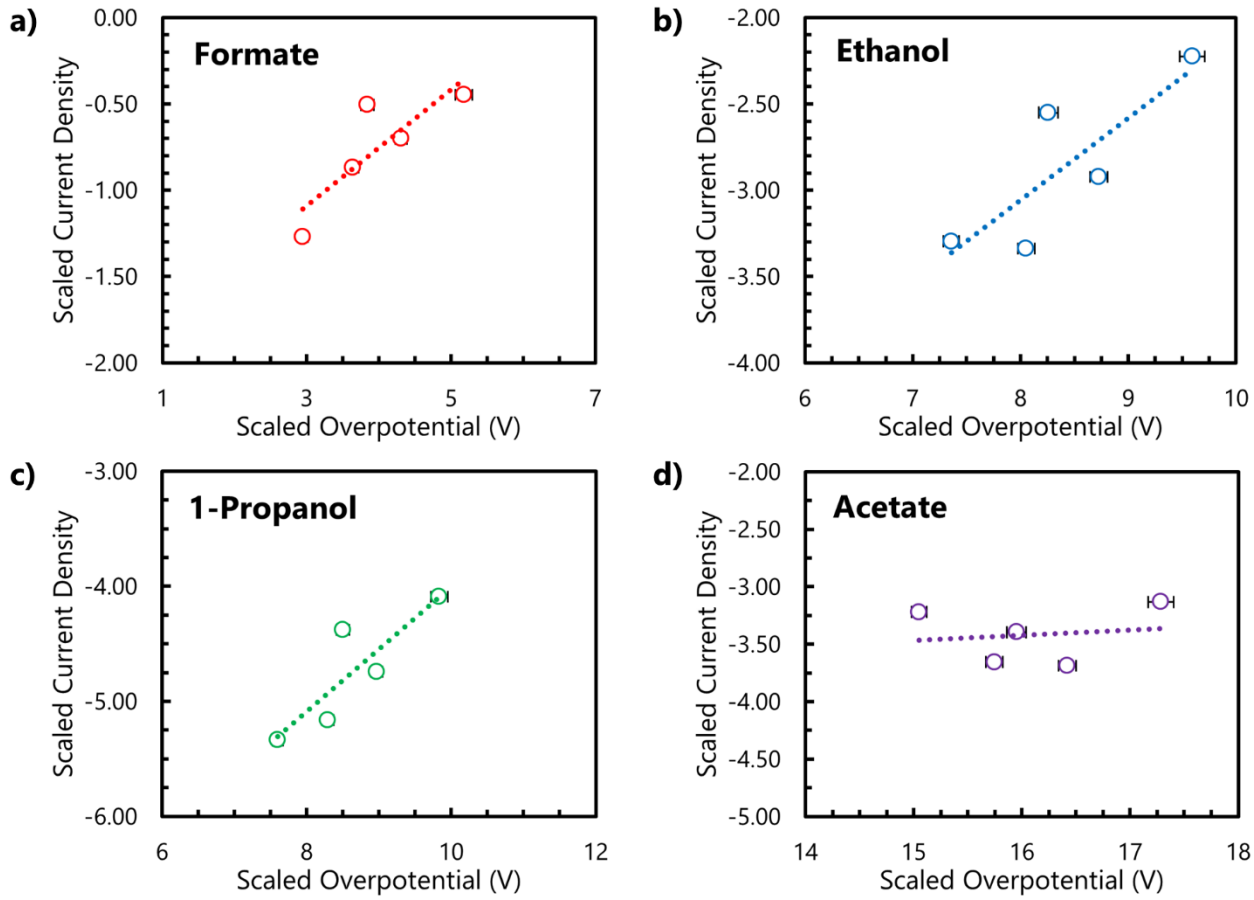

Figure S8. Electrochemical kinetics for liquid products of CO_2 electrolysis. (a) Formate. (b) Ethanol. (c) 1-propanol. (d) Acetate. The values plotted were derived using the lab-scale CO_2 electrolysis data at five different currents: 25, 50, 75, 100, and 150 mA cm^{-2}. Horizontal error bars correspond to the standard deviation of the cell voltage over the final 15 minutes of operation.

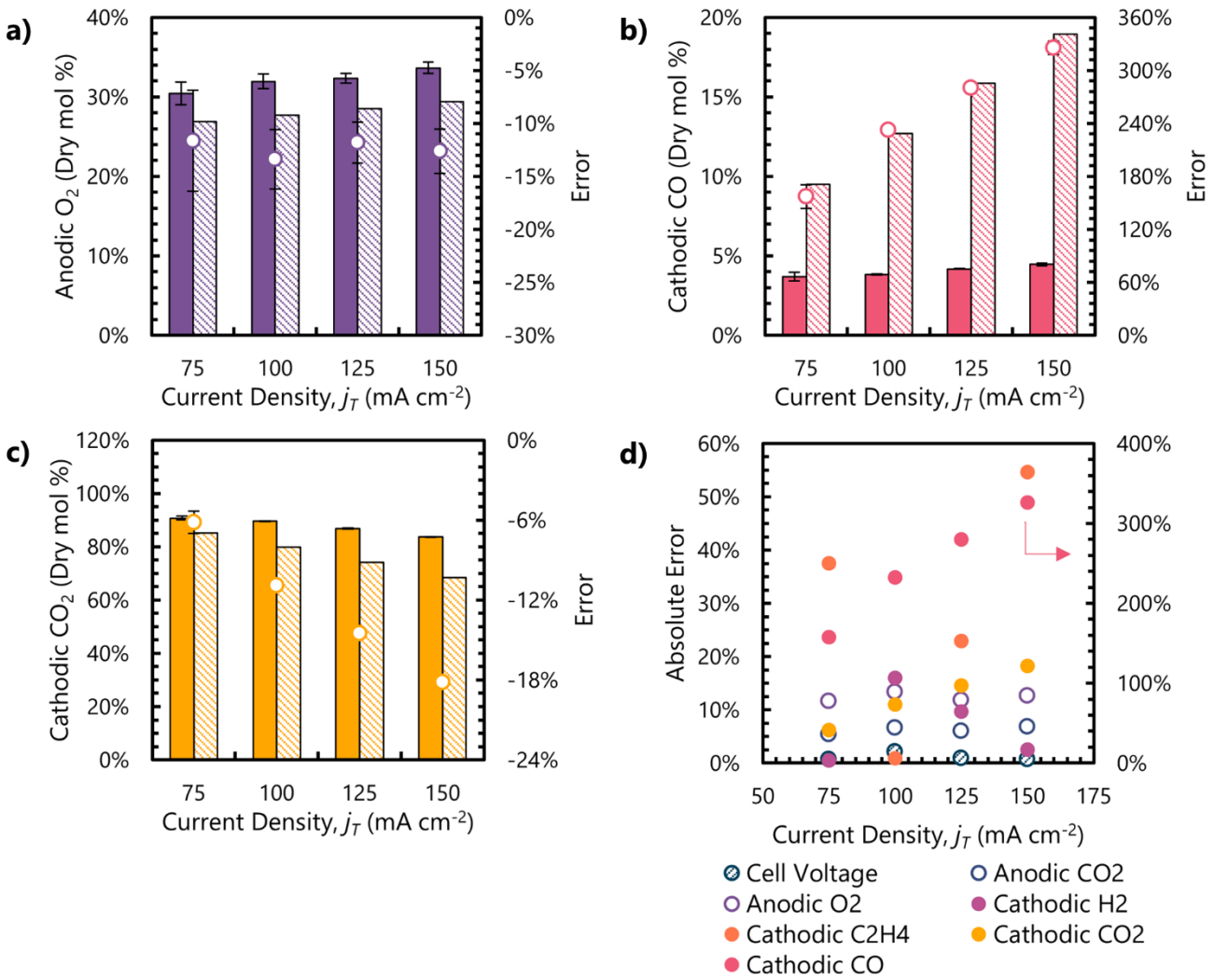

Figure S9. Model validation against 800 cm^2 pilot-scale single cell. The experimental results and model predictions are marked with solid and patterned fill, respectively. (a) O_2 in anode tail gas. (b) CO in cathode tail gas. (c) CO_2 in cathode tail gas. Error bars correspond to the standard deviation of three independent measurements. (d) Absolute error of the seven performance metrics at different current densities (error bars omitted for clarity).

CO/CO_2 Concentration Discrepancy. The model predicted a much higher CO concentration than was measured in the experiment. Converting the concentrations into a FE value could help to understand the differences between the model and the experiment. Both the flow rate and the composition of the cathode tail gas are required to determine the FE towards a specific product but the cathode flow rate at the pilot-scale was not measured. The flow rate leaving the cathode depends on the CO_2 supplied, the CO_2 converted, gaseous products generated, and CO_2 crossover to the anode. Since the model demonstrated an effective prediction of CO_2 crossover (**Figure 3b**), it is likely that the model accurately predicted the CO_2 which remained on the cathode (i.e., that which is converted and unreacted). An approximation was employed to estimate the FE of CO and C_2H_4 assuming that the FE towards H_2 predicted by the model, denoted $FE_{H2,Model}$, was accurate (e.g., matched that of the experiment). To compare against the FE predicted by the model, the experimental FE towards product i, denoted $FE_{i,Exp}$, was calculated using the following formula:

$$FE_{i,Exp} = \frac{z_i C_{i,Exp}}{z_{H2} C_{H2,Exp}} FE_{H2,Model} \tag{S29}$$

where z_i denotes the number of electrons transferred per mol of product i, $C_{i,Exp}$ denotes the experimental molar fraction of product i in the dry cathode tail gas, z_{H2} denotes the number of electrons transferred per mol of H_2, and $C_{H2,Exp}$ denotes the experimental molar fraction of H_2 in the dry cathode tail gas.

The combined FE of the gas products (H_2, CO, C_2H_4) from the experiments was estimated at 59% at 75 mA cm^{-2} and fell to 48% at 150 mA cm^{-2} (**Figure S10a**). By comparison, the model predicted that gas products should have a combined FE of 86 – 87% when operating over the same range of current densities (**Figure S10a**). The combined FE predicted by the model was similar to the lab-scale experiments at the same current densities (**Figure 4a**). At 150 mA cm^{-2}, the model predicted an FE_{CO} 32% greater than the experiment results. Based on the reduced combined FE

towards gas products, the experiment seems to have generated more liquid products, possibly due to differences in mass transport in the larger cell.

The model's overestimation of the CO selectivity resulted in a significant underestimation of the CO_2 concentration in the cathode tail gas – reaching as high as 18 mol% less at 150 mA cm^{-2} (**Figure S9c**). When CO_2 is converted into gaseous products, CO_2 molecules are removed from the gas stream and product molecules are inserted, thereby lowering the CO_2 concentration in two ways. Liquid products impact the CO_2 concentration in the cathode tail gas less since they only remove CO_2 and do not introduce new species into the gas stream.

To demonstrate the sensitivity of the CO_2 concentration on the liquid product selectivity, a sensitivity analysis was performed using the electrolyzer model (**Figure S10b**). At 150 mA cm^{-2}, the model predicted the that $HCOO^-$ and C_2H_5OH were the most produced liquid products at 9% and 3% FE, respectively. FE_{CO} was shifted to/from $HCOO^-$ or C_2H_5OH (individually) while all other products retained the originally predicted FE values. If FE_{CO} was reduced by ~30% relative to the model's initial prediction (suggested by the experimental results in **Figure S10a**) then the CO_2 molar fraction in the cathode tail gas increased by 12-14% over the base case model prediction (hollow marker in **Figure S10b**), closely matching the experimental results. Producing $HCOO^-$ instead of C_2H_5OH yields lower CO_2 concentrations because more CO_2 is converted (i.e., removed from the gas stream) on a charge-normalized basis, one CO_2 molecule converted per 2 electrons *vs.* one CO_2 molecule converted per 6 electrons, respectively.

To confirm that the pilot-scale experiment produced additional liquid products instead of CO, liquid products should be collected and the gas flow rates should be measured to enable the direct quantification of experimental gas and liquid FE values.

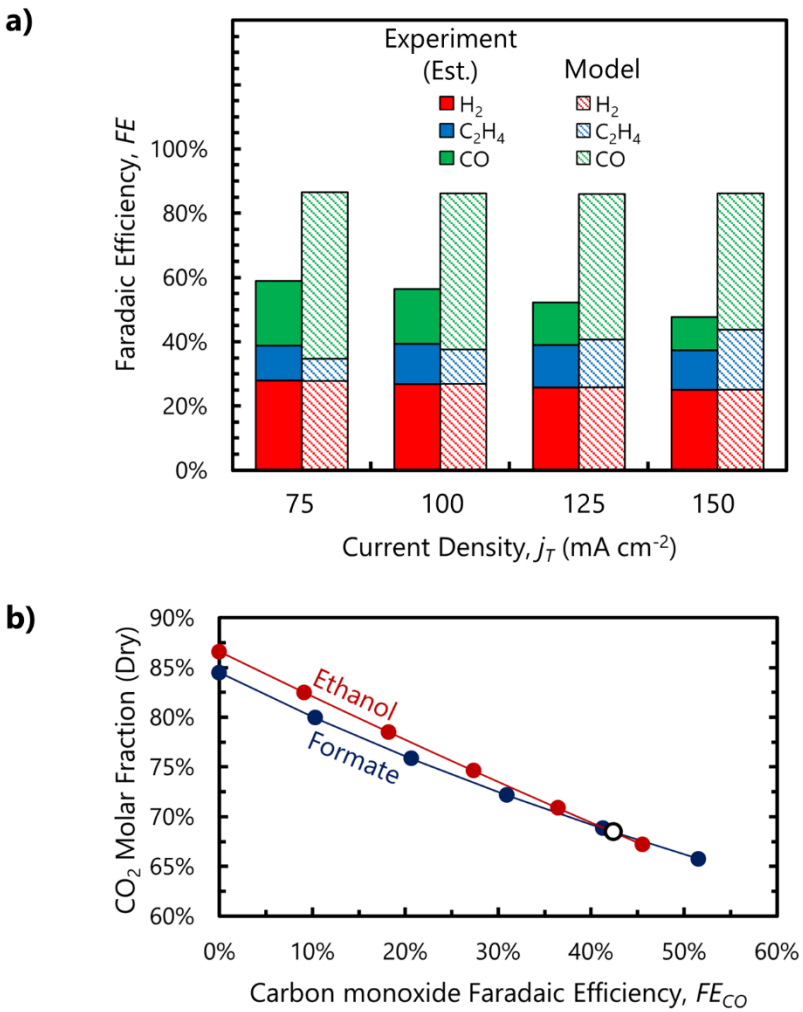

Figure S10. Understanding the discrepancy in cathodic CO concentration for the pilot-scale electrolyzer (single 800 cm^2 cell). (a) Faradaic efficiency (FE) towards gas products from the experiment and the model. The experimental FE values were estimated using (S29) and were not measured experimentally. (b) Sensitivity of the CO_2 concentration in the cathode tail gas to the CO-liquid product distribution (modelled at 150 mA cm^{-2}). FE_{CO} was translated to/from a single other liquid product (formate or ethanol). The hollow marker indicates the FE predicted by the model using the kinetic constants defined in **Table S4**.

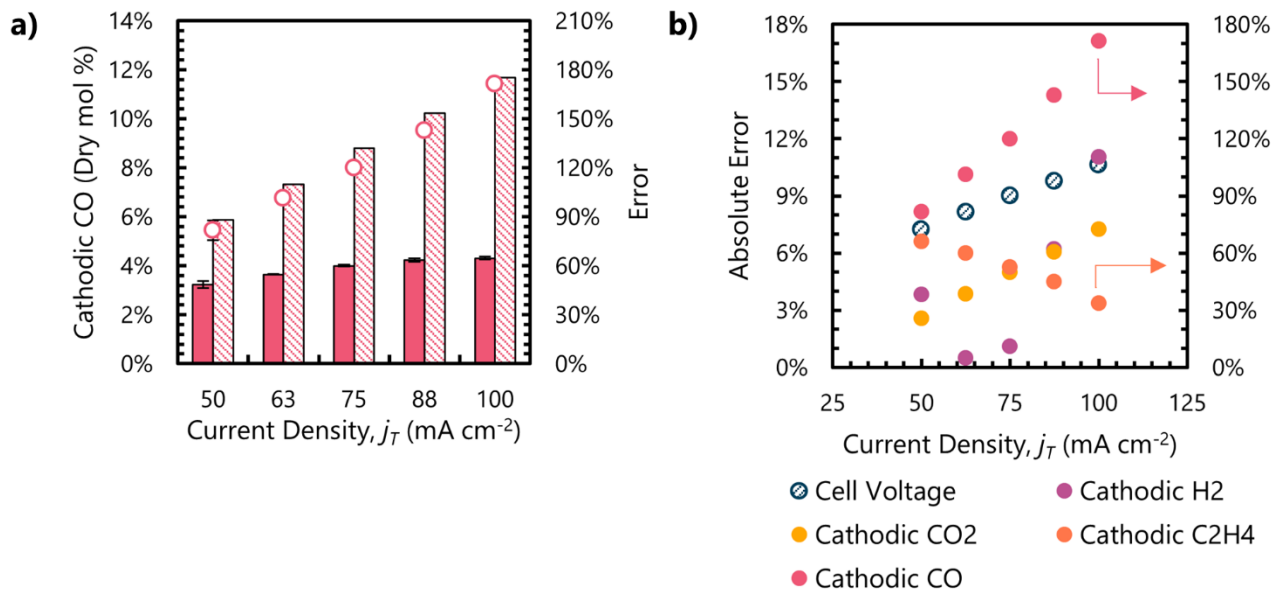

Figure S11. Model validation against 10 x 800 cm^2 pilot-scale stack. The experimental results and model predictions are marked with solid and patterned fill, respectively. (a) CO in cathode tail gas. Error bars correspond to the standard deviation of three independent measurements. (b) Absolute error of the five performance metrics at different current densities (error bars omitted for clarity).

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